

Models for CALL (5. Models for CALL. 6. CALL Models Practices in SLA)

This **traditional model** by **Dunkin and Biddle (1974)** is examined in terms of how it can and cannot accommodate CALL. Based on the differences between **CALL** and traditional language learning, a new model for learning with CALL is offered. **Collaboration** is explored in terms of the challenges raised by learners themselves (**social challenges**) and challenges specific to the computer and CALL programs (**technical challenges**).

The principal concerns of this lesson can be summarized by the following **three** questions.

Model of CALL

The introduction of the computer into the classroom (in some cases, some places) has fostered changes to the practice of the **teaching and learning** of languages and related subjects, such as literature. Of particular interest to CALL research is the ability of **multimedia** to increase comprehensible input and offer opportunities for comprehensible output thus fostering **SLA**, through **collaboration**, which was a key construct in understanding opportunities for learning with CALL. In order to elucidate the relationship between CALL and SLA, it is necessary to construct a **model**. A model is a **pictorial representation of a metaphor**; a way of looking at and understanding the world. A model can be used as a tool to examine processes and describe the ways in which **teaching and learning** may take place or may be improved upon.

As CALL develops and increases in **complexity**, new models are put forward to describe more **discrete portions** of the discipline. For example, **Bertin and Narcy-Combes (2007)** focus on the role of the learner in a **computer-mediated environment**, suggesting that monitoring learner outcomes can be counter-productive.

Dunkin and Biddle (1974) noted that there appears to be a never-ending supply of models but go on to suggest that there are good reasons for such being the case, ‘**teaching** is a complex activity that reflects many factors.

Problems in Creating CALL Models

One is the **expansive scope** of what is considered to be within the realm of CALL processes, especially considering the broad definition for CALL that takes into consideration its changing nature: any process in which **a learner uses a computer and**, as a result, improves his or her language. A model for CALL either has to accommodate this general definition or more narrowly define a set of **significant aspects** of CALL that might be considered. The model offered later in this lesson attempts to define CALL in terms of classroom practice rather than also deal with learners using CALL software in other contexts, such as at home or as an activity **peripheral** to another subject.

The Need for a CALL Model

The question of whether or not CALL is **intrinsically** different from traditional language teaching and learning has been addressed through an examination of several **historically innovative software programs**. The fact that CALL is different from **prior approaches** and materials is also pointed out by **Williams (1998)** who argues that the potential for teaching and learning through CALL has not been tapped.

N. Williams emphasized upon the need for reviews of the particular nature of CALL technology and proposed that; “Hypermedia allows everything to be linked to everything, just like human imagination. But teaching (as opposed to learning) requires rather narrower perspectives. Whilst the information within educational multimedia is extensive, attractive and complex, the interactions with and within that information, allowed by the authors of such systems, fail to realize the educational potential of the medium. We need more radical reviews of what learning might be through multimedia, combined with innovation in design (Stringer, 1997), to realize more fully the true potential for educational interaction in multimedia”.

Creating a new model requires decisions on the range and types of variables that need to be considered. The first task in describing a new model for CALL is to identify a current model for teaching and learning and examine what variables might or might not need to be reassessed. Hubbard (2005), in a review of 78 articles published in CALL journals between 2000 and 2003, cautions that most of what we know about CALL seems to come from studies conducted with untrained novice users.

A Model of Current Non-CALL Language Learning

Many authors have attempted to isolate the variables involved in learning and teaching language. Spolsky (1987), for example, lists 74 variables affecting learning. Such a high number of variables makes it difficult to isolate any one variable for careful consideration in, for example, an experimental examination of the variables’ influence on learning. A general model that includes and expands on Breen’s (1998) criteria can also include observations from Dunkin and Biddle’s (1974) model. Their model also identified aspects of language learning that might be studied in a classroom.

Dunkin and Biddle’s model is a general one, in the sense that it covers all formal instruction, not just language teaching. The following sections examine Dunkin and Biddle’s model in terms of its ability to accommodate CALL and outlines the differences that help define a new model suitable for CALL.

Various Dimensions

There are two ways of considering CALL: either as a supplementary learning material in which case CALL simply falls under the Classroom contexts heading in Dunkin and Biddle’s model; or as an autonomous process, a kind of virtual teacher. there is so far no empirical evidence of a learner completely acquiring a second language through CALL alone, although this is an interesting area for a study; the task would be to create or assemble a comprehensive CALL software program(s) for a language unknown to the teacher, the learner and the community, such as Swahili in rural Saskatchewan. Instead, it is assumed that the role of CALL is to help foster language learning by creating conditions that make some aspects of language learning easier.

Computer Functions beyond Traditional Materials

However, it is also necessary to accept that the computer sometimes assumes functions beyond traditional materials and accept that its inter-activity mirrors at least some of the functions of the teacher, especially when it is used autonomously, even though it cannot pretend to duplicate the teacher’s range. It is with this view of CALL that Dunkin and Biddle’s model is examined.

In Dunkin and Biddle's model, there are aspects that are broadly fixed in both **non-CALL environments** and **CALL environments**:

- **School, community and classroom contexts**
- **Product variables of immediate pupil growth and long-term pupil effects**

School, community and classroom contexts are likely to change only in the sense that some learners presumably encounter computers and computer software well ahead of their introduction into the classroom; or through the adoption of a **virtual** classroom model. This view, as already mentioned, needs to assume that CALL materials are more than just a classroom con-text. Through exposure to computers and computer software, learners are prepared for the ensuing classroom variables; in some cases, learners are given insight into how teaching materials are constructed – what Dunkin and Biddle consider the Presage variables, which we will discuss in coming topic.

The other factor unlikely to undergo significant change is Dunkin and Biddle's Product variables of **Immediate** pupil growth and **Long-term** pupil effects, although the ways of measuring such changes, both by the teacher and the learner, are likely to evolve, for example, through **computer-adaptive** testing programs in which the computer evaluates a learner's early performance in **subtests** and builds fuller tests, as well as **scoring and tracking** progress through learning programs.

Presage Variables

Presage variables include Teacher formative experiences, Teacher training experiences and Teaching skills. **Software** programs might in some cases, with different emphasis, function as a virtual teacher, **a teacher's helper, a teaching tool and a learning tool**. Sloane (1990) as well as Boyd and Mitchell (1992) take a discourse approach and consider CALL a conversational activity shared among the learner, **the machine tutor** (i.e. computer program) and the **educator/developer**. But in many CALL programs, Dunkin and Biddle's teacher attributes are generally subsumed under the consideration of the program taking the **place of the teacher**.

Dunkin and Biddle (1974) assume that an **individual** teacher comes to the classroom with a wide range of **experiences and skills**. Teacher **formative** experiences refer to factors such as those experienced because of social class and sex. They also list age, race and **physical appearance** to illustrate the fact that everything about a teacher is inclined to be involved in how the teacher is 'likely to be treated differently both within and without the school' (p. 39). In terms of a CALL context, the formative experiences are likely to be **homogenized** into the collective experiences of the materials developers as many **soft-ware developers'** marketing executives are likely to be involved in any decisions in the creation of a CALL program.

Teacher training experiences refer to the formal education that a teacher receives from both **academic institutions** (i.e. **teacher colleges, universities**) and training in **school placements**. Dunkin and Biddle (1974) note that these include 'courses taken, the attitudes of instructors, experiences during practice teaching and **in-service** postgraduate education, if any' (p. 39). **Hampel (2009)** defines some of the skills necessary for teaching in a virtual environment, but in the CALL context, again, because of the collective

nature of the creation of CALL materials, these experiences are likely to be **homogenized** into particular ideas of models of instruction. There is room for learners and teachers to organize the ways in which they learn, but these are often limited by what **software** developers and other personnel involved in the process will agree on in terms of organizing the approach to learning based on their perception of what is an appropriate model.

Teacher skills refers **to psychological traits, motives, abilities and attitudes** and rely on what Dunkin and Biddle (1974) note as ‘the common view that teaching is largely a matter of personal relationships and personality, that such effects as teachers have are functions of their personalities (Nuthall, 1972, personal communication)’ (p. 40). How a CALL context might have a ‘**personality**’ is difficult to assess, although **Microsoft Windows and Apple Macintosh** interfaces are often characterized as each having a different ‘**feel**’ and there are many aspects of an interface which can be measured in degrees of user-friendliness.

In a CALL environment, the feel or **user-friendliness** of the program is often determined by the affordances of the program (i.e. what the program is or appears to be capable of doing in terms of both intended and unintended functions) and how easy or difficult it is to understand these **affordances**. In general, the trend in CALL has been always to make **interfaces and the information** within them as easy to understand as possible. However, it may be that this does not reflect good teaching practice or real-life situations in which teachers and other interlocutors are likely to present information in obscure or veiled ways to encourage thinking on the part of the learner. As CALL programs continue to have a high degree of interactivity with both teachers and learners, it is likely that learners will feel less isolated by program features that do not seem to address their particular needs. The **continuing high quality** of both **animation and online video** present opportunities to challenge learners to think, particularly when the teacher can make an appearance in the form of an animated or video avatar.

But many CALL programs continue **implicitly** to state ‘I have the answers to your questions; It is difficult for computers to deal with **ambiguous learner** input, but this is an area of research that needs to be further investigated.

(b) Context Variables

Dunkin and Biddle’s context variables of **Pupil** formative experiences and Pupil properties are different in a CALL environment because many learners are likely at least to come into contact with computers, if not CALL, before coming to school. In this way, they are **pre-socialized** into the idea and methods of learning with computers and develop their own **competencies** and learning strategies. Learners who bring computer and CALL formative experiences and properties to the classroom assist in defining the variables of a new model of learning because they see using the computer as something that is distinctly different from traditional classroom instruction.

Such transfer – or **lack of transfer** – may limit the variables of a new model because the learner may be unable or **unwilling** to accept new learning ideas and practices. This has certainly been the case with a generation of people who have pronounced themselves **computer-phobic**; but such a **phobia** is generally unknown among young children who have grown up with computers and take them for granted. **Young** learners are more willing to accept new models of learning because they are not conditioned into regarding the existing model (as described by Dunkin and Biddle, 1974) as a so-called ‘**right**’ or ‘**normal**’

way of doing things. The computer's exciting yet controlled presentation of learning materials is attractive to many young learners.

Morariu suggested factors for the design of the learning environment

Depending on one's vantage point, these factors can shift. Morariu (1988) suggests **four** other factors from the point of view of the design of the learning environment that are not **explicit** in Dunkin and Biddle's model.

Morariu (1988) suggests that CALL materials' developers should decide where in **Bloom's taxonomy** of questions, learners should be operating. When learners take in information, they use various strategies to make sense of it. These strategies may correspond to levels of questions or thinking outlined by Bloom (1956) in his Taxonomy of Questions. These questions range from **low-level** knowledge and factual levels to higher levels of analysis and evaluation. Strategies such as skimming and scanning allow learners to answer lower-level questions while **higher-level** strategies, such as reading for attitude, allow learners to answer higher-level questions and, generally, to think in greater depth about a text.

J. Morariu commented on the design of learning environments as:

- **Goals/Objectives.** Stated in behavioral terms, a full breakdown of the con-text and measurable outcomes for the entire instructional environment.
- **Navigation.** The user interface design that defines how the learner can move through the system (e.g. Are predetermined 'tours' provided? How does the learner know where he/she is? Can the path be retraced easily? Are graphic icons used for browsing /selecting information or do users need to type key words?)
- **Structure.** The overall organization of the information (e.g. hierarchical with topics and sub-topics, associative with word/icon links).
- **Format.** Media for presenting the content /data (e.g. text, graphics, animation, audio, still images, motion video).

Morariu (1988: 18–19)

Bloom's taxonomy was revised by Anderson and Krathwohl (2001), changing the category titles to verbs and shifting the focus of the **upper-level** items to: **remember, understand, apply, analyze, evaluate, create.**

In a CALL program, it is important to have **information** structured on a series of levels encouraging readers operating at various levels, at any point, to delve deeper into explanations of the content. It is also important that programs challenge students to perform at higher levels within **Bloom's (1956)** taxonomy. For example, a program presenting vocabulary for shopping might allow learners to follow hyperlinks to delve deeper into more complex expressions and vocabulary. Such a program might also ask learners to recall information and to apply what they have learned to new situations and generate their own rules about the **grammatical** rules that are involved.

Teacher and pupil classroom behaviour: activities used in CALL

Computers inherently allow for a greater learner autonomy because, unlike a teacher, they are available beyond the time and space confines of the classroom; a learner who wishes to revisit and extend his or her learning at any time or place (that is, any place with a computer) has a potentially rich resource with which to do so. CALL resources are extremely limited in some ways when compared to a well-qualified teacher but, in most cases they far exceed what is available from a textbook.

The range of tasks and exercises available in CALL can be organized into various taxonomies based on the stated focus of the software (e.g. grammar, vocabulary, fluency), targeted language skills (e.g. reading, writing, speaking and/or listening) or Bloom's (1956) levels of questions and learner characteristics based on age, gender and level (e.g. beginner, intermediate, advanced). But a more general way of visualizing what goes on at the computer can be based on the degree of involvement with which learners access information. This measure can be seen within a continuum of locus of control.

(a) Locus of Control

Locus of control refers to the continuum between the program's and the learner's responsibility for decisions about the sequence and outcomes of learning, learner interactions and, even, the content of what is taught. Jonnassen et al. (1993) see the distal ends of the continuum as representing objectivist and constructivist orientations towards learning. Instead, emphasis rests on individualistic and competitive goals. Simulation games and Experimental simulations are more likely to fall into a constructivist model of instruction that acknowledges that learning best takes place when learners struggle with information while trying to fit it into a meaningful schema.

Chandler's categories of content-free tools and programming languages may be seen as neutral in terms of a model, although they are likely to be combined with tasks and used in a classroom in a constructivist approach to learning.

Perhaps too late for Chandler's model is the increasingly popular use of the Internet for various forms of learning and discourse, summarized by Turkle (1995) and Seaton (1993).

(b) Observable Changes in Pupil Behavior

A good example is found in Ward et al. (2007) on providing opportunities for learners to model back-channel feedback (expressions such as 'uhhuh' that demonstrate comprehension when listening) in an Arabic program.

(c) Measurement of Language Learning

Traditional measurement of language learning depends on teacher observations and tests. But CALL programs present many opportunities for different kinds of measurement. For example, a learner's links can be tracked to see the extent of a program a learner explores and the time he or she spends in each section. More importantly, many CALL programs allow for a learner to obtain immediate feedback on progress, that is, an opportunity to observe their own changes in behaviour. This continuous formative assessment differs from a traditional classroom teacher's random spot-check of learners' comprehension or a final test.

Definition of Virtual Classroom

A **virtual** classroom is an **online** classroom that allows participants to **communicate** with one another, view presentations or **videos, interact with other participants, and engage with resources in work groups**.

(a) Characteristics of a Virtual Classroom

1. Non-Restricting

A virtual classroom allows both learners and instructors around the world to participate in live classes to collaborate and interact. MOOC programs like Coursera are a great example of this concept in action.

2. Affordable

The low costs of virtual classrooms are considered to be a major advantage. Learners can save money by not having to worry about travel expenses. Participants also save time since all that is needed is an internet connection.

3. Flexible Learning

Online classes also allow for the ability to record class as it happens, including any presentation audio and visuals. This means that the content is accessible even after being delivered, an added benefit for those who want a quick refresher, or perhaps did not fully understand the first time.

4. Practical and Proven

Synchronous learning is a learning environment where everyone takes part in the learning at the same time. A traditional lecture is an example of this type of learning, and has been used for hundreds of years. Online learning enables this same type of experience, but with far more conveniences and tools.

5. Accessible

Virtual classrooms can be used to deliver lectures, or even tutorials online. They are also great options for impromptu meetings and group projects where members need to check-in on progress and bounce ideas off one-another. With the virtual environment, ideas and collaborators are never far away.

How Virtual Classroom Works?

Wesley and Franks (1994) suggest the virtual classroom class is ‘an electronic classroom which can be expandable in **time, space and content**. Its **informational territory** can grow indefinitely as new knowledge and resources are acquired and as the capabilities of new members are added’ (p. 3). They also suggest that the virtual classroom has continuity through time in that it is not limited to conventional academic time segments (**semesters, school years, etc.**). As **successive groups** of learners are added, previous learners need not leave, but can remain to continue their learning and to support the learning of the new students. In the intervening years since Wesley and Franks defined the virtual classroom, it has taken on a range of meanings from the fully **animated avatars** of virtual **Second Life** campuses, to online **Blackboard** teaching modules that also provide live chat functions.

Most of these thoughts focus on a distance-learning model using email and the WWW for **computer-mediated communication (CMC)**. However, most CALL materials work in similar ways.

Aspects of a CALL model

Although Dunkin and Biddle (1974) refer to challenges among teacher proprieties such as **authoritarianism and anxiety**, the above adaptation to a CALL context does not address all the challenges in the learning process but serves as a starting point to examine what goes on in CALL contexts.

Association of CALL models with CALL Process

To understand these, various views of CALL were presented along with a **summary** of activities used in CALL and an overview offered in terms of a model of **locus of control at the** computer. This model suggests that activities in CALL can range from little or no control on the part of the learner to complete control. It was suggested that the former signals a **behaviourist** approach while the latter falls into a constructivist approach.

CALL Material Problems

CALL materials initially suffered from **peripheral** problems, including a **lack of funds, a lack of expertise and a lack of suitable tools** in the form of authoring programs that in turn create problems with defining a suitable model. However, **user-friendly** online applications are increasingly used to create materials which, although incapable of offering a comprehensive language-learning program, nonetheless fill some gaps that complement the work of a good teacher. A CALL model that can help guide materials developers and those who assess CALL materials is a **noble** pursuit, but in the fractured approaches to teaching and learning a language at the computer, it is unlikely a practical model will encompass everything being done in a **meaningful way**.

CALL Applications (7. CALL Applications I. 8. CALL- Language Games 9. CALL Applications II)

CALL Applications

Simulations are ideal for offering **immersive** environments that offer exposure to the target language in various **media** but sometimes lack the focus of other language teaching approaches.

Word Processing

Almost all computers are sold with some version of word processing already installed, usually **Microsoft Word**. Such programs are widely used in the composition process. Within such **wordprocessing packages**, **spelling and grammar checkers and simple thesaurus programs are standard tools**.

There used to be only one way for learners to correct spelling on their own: using a **dictionary**. Now, as learners increasingly use computers in the composition process, they frequently do their spelling corrections with **computer-based spelling** checkers and seldom refer to a dictionary or use an **online dictionary**. **Spelling** checkers are also included in other applications such as database programs because as many as one in **ten data-base** queries fail because of spelling mistakes either in the query or within the database index itself (Smith, 1991).

After much research in the 1980s and early 1990s, attention has shifted away from the influence of spelling checkers and grammar checkers. However, it is an area which continues to merit attention as learners turn away at younger ages from writing on paper to computer-based composition.

Although computer software manufacturers may consult educators, most word-processing programs and other applications are designed not for school use, with attendant pedagogical concerns, but for business environments where learning is less important, or even completely unimportant. In business, the focus is on the completion of tasks. This is particularly seen in programs that offer spelling correction but do not provide any definitions. Learners – particularly second-language learners – frequently misspell a word then choose the first correction offered, without considering whether it is appropriate or not. Moreover, word-processing programs do not record misspellings or give any feedback to learners interested in reviewing their errors. Even worse, spelling checker programs allow learners to add new words to standard dictionaries and sometimes these words are themselves misspellings.

It may be that spelling checkers assist in fluency by relieving learners of undue concern over their spelling during the composition process

Word-processing programs also provide grammar support, although such support is sometimes of questionable worth. The grammar checker included with various versions of Microsoft Word, for example, is pre-programmed to object to the use of the passive voice, even when the passive voice is appropriate. This and other rules can be suppressed within the program but it may be difficult for learners to do so and would require that they have a certain level of grammatical sophistication to appreciate why it might be done.

Beyond word processing, software such as Microsoft Word is increasingly multi-purpose. For example, it is commonly used by teachers (and some-times learners) for creating semi-authentic learning materials featuring text, tables and illustrations as well as simple websites.

Games

Most educational games, or games used for pedagogical purposes, make use of a form of subversive teaching; learners are unaware of the objectives or, rather, do not share the same objectives as the teacher. Instead, learning takes place as an activity peripheral to play. Learners asked to play the board game Monopoly, for example, may not be aware that they are learning rudimentary concepts about the value of money and real estate, nor will they associate these concepts as part of learning objectives. The peripheral learning benefits in a game are likely to be small but are hopefully greater in a program devoted to some specific educational objective. The best educational games are those which embed the pedagogical objectives so that the learners' perceptions are of play, while the teachers' hidden objectives are still achieved.

In other cases, learning materials which are not perceived as being game-like by the teacher may be perceived as such by the learner. For example, Nord describes an exchange with his students at Nanzan University in Nagoya, Japan in which they asked for another session in the ComputerAssisted Language Learning (CALL) Laboratory using a particular game.

Among computers' most popular uses today are as platforms for arcade-style computer games although an interesting development has been the introduction of the Nintendo Wii gaming system which employs

kinaesthetic skills in doing games; it is a platform rich for development of educational applications, particularly for younger children. Games are extremely attractive to young learners (and many older ones) and those games that foster collaboration can provide opportunities for **scaffolded** language learning (Papastergiou, 2009).

A degree of excitement is necessary in **computer-based learning** materials, particularly in the case of young learners who lack motivation for learning. Within the classroom environment, computers can help in motivation through the organization of learning into **game-like** formats.

On the simplest level, the **computer** is a suitable game player as it can provide clues, levels of difficulty and rewards for solutions through points or visual stimulation. Most importantly, the computer is endlessly **patient and never grows bored**. Simple computer games include variations of many of those found in the **real** world.

Commercial Quiz Software

Particularly online, computer-based games are often in the form of **quizzes** which test knowledge more than they teach it. Like **games, quizzes** are very motivating for learners as they appear to illustrate a learner's progress and give some security against fear of more **formal** exams. As **computer-based** quizzes are often done outside of class and not marked by teachers, learners may feel less threatened.

A popular quiz format is a **cloze** exercise, such as one of several test types found in Respondus (<http://www.respondus.com>), a popular assessment tool used with computer-based course design packages such as **Blackboard Vista**. **Deleted words** are replaced with numbered blanks. Learners can request clues in the form of **single** letters. Such programs encourage careful reading but frustration may occur as the programs do not accept **synonyms**; usually learners must guess the exact word that has been deleted.

Other quiz software includes programs to check **spelling, listening, speed reading, knowledge of synonyms and antonyms, general knowledge and other English skills**.

Literature

One of the ways in which learners are presented with opportunities to acquire language is through activities peripheral to the study of literature and other **disciplines**. A work of **literature** is not a simulation, but it has a high degree of **fidelity, or authenticity**, in that the learning materials are both extensive and taken from real-world sources. **Literature** forms the basis of many CALL programs.

Collie and Slater (1987) offer four main justifications for the use of literature in the language classroom: **valuable authentic material, cultural enrichment, language enrichment and personal involvement**. They also note the place of **non-fictional** authentic materials, such as bus schedules.

Literature is a valuable complement to (other classroom) materials, especially once the initial '**survival**' level has been passed. In reading **literary** texts, students have also to cope with language intended for **native** speakers and thus they gain additional familiarity with many different **linguistic** uses, forms and conventions, and so on.

A **rich** source of content for CALL programs is literature which has come into the public domain and is freely available in digital formats, often with accompanying visual materials. This allows learners to manipulate **text and comment** on it in interesting ways. **Farabaugh (2007)**, for example, talks about how students can use wikis to make sense of a **Shakespeare** play: commenting, editing comments and generally building their own knowledge base with personal opinion and other sources.

Skills in making **inferences and deductions** can be taught in different ways, but literature tends to teach them more naturally.

The relationship of literature to computer-based learning materials

Collie and Slater's (1987) observations on the link between language learning and literature were written with reference to traditional paper-based learning materials they also hold true for computer-based learning materials. However, there is a principal difference: **computer-based** learning materials easily bridge the gap between fictional and non-fictional resources by routinely offering multimedia links between the two. In **paper-based materials**, this is sometimes found to a lesser degree in heavily annotated editions of literary works such as Shakespeare plays. A **Shakespeare** play presented on the computer is likely to offer video of the play being performed, diagrams of costumes, virtual tours of the original Globe Theatre, **interactive** dictionaries and other tools for literary and language enrichment.

Hypertext and Literature

Deegan and Sutherland (1990) note advantages of hypertext for both the study and writing of literature, arguing that **texts and non-textual** material are essentially fluid and easily manipulated, making it an ideal tool for showing the interconnection of ideas.

Landow on the advantages of hypertext as a tool for learning connectiveness in literature:

The sheep like behavior displayed by many freshmen is often due to their having little information and little idea of what to do with it. One cannot make connections between fact A and six other facts if one knows only fact A. This lack of factual knowledge leads to reductive thinking. Additional information, however, will not help students think critically unless they have techniques for relating facts to each other. College liberates because it provides students with facts and offers examples of the way they can make connections for themselves. Intellectual freedom derives from an ability to make choices. Anything that can help teachers communicate information to students as well as provide them with techniques to relate it to what they already know provides a model for education. The habits of mind, thus encouraged, apply to all kinds of activities, inside the classroom and out, and they remind us that education and thinking are active procedures.

Landow (1989: 176)

Corpus Linguistics

Corpus linguistics is an important area in its own right within applied linguistics, but it is also a useful tool for the teaching and learning of language at the computer.

Using **corpora** in the classroom involves making use of a concordancing program's ability to spot patterns and exceptions in language use.

Corpus Linguistics in the Classroom Both teachers and learners can use corpus linguistics in various ways within the classroom.

A teacher might collect a set of student assignments and use a concordancing program to analyse examples of learners' language looking for typical error patterns. Systematic errors in learners' writing can be used as a basis for the development of learning materials. Learners themselves can be trained in the use of a concordancing program (St John, 2001) and corpora, then become their own researchers finding examples and developing their own rules for grammatical structures, idioms and general usage, for example, investigating the differences between the verbs look and see. This approach is often called Data Driven Learning (DDL).

Computer-mediated Communication

Communicating using the computer is often referred to as computer-mediated communication (CMC) and is one of the more popular activities associated with CALL. CMC encompasses communication by email, bulletin boards, chat lines, within MOO (Multi-user domains, Object Oriented) environments and using social networking services such as Facebook and Twitter.

CMC refers to a situation in which computer-based discussion may take place but without necessarily involving learning. Of course, opportunities for learning are inherently present, especially in situations in which second-language learners need to engage in negotiation of meaning with native speakers of the target language or even with peers of non-native proficiency.

Email

Email is among the most popular uses of the Internet and presents many opportunities to enhance learning.

Using email, learners can communicate with peers, teachers and native speakers. Messages can be structured around an assignment in which the learner solicits special information, shares information about assignments (especially in a jigsaw format, sharing information with peers) or submits thoughts, questions and assignments to a teacher. However, communication with native speakers can be difficult if the learner provides input with substantial spelling and grammatical errors. Most email programs now come with rudimentary spelling checkers but teachers concerned about the quality of their students' writing may encourage them to compose their messages in a word-processing software program then copy and paste it into their email messages.

Net Pals

One informal use of email is for the establishment of email pen pals. Such pen pals or pen friends are sometimes called net pals. Ideally, net pal communication is between someone learning the target language and a native speaker of the target language living in the culture of the target language. For example, a Nepalese student learning German might correspond with a native German-speaker living in Berlin. The advantage to the learner is that the native speaker is likely to offer extensive examples of authentic language, probably pitched at the appropriate age level of the language learner.

In practice, however, both parties can find net pal relationships **frustrating**. The language learner may find that **slang, idioms, misspellings** and typos interfere with comprehension. Net pals work best when both parties have **extrinsic motivation (e.g., a common interest in football or visiting one another's countries)** and where the language learner has good clarification strategies and/or the support of a language teacher.

Chat Lines

A **chat line** is more technically known as **Internet Relay Chat (IRC)** and appears on-screen as a window that presents what the learner is writing in one pane while general discussion among other participants continues in another. Once the learner has completed a message and presses the **send command**, the message is queued and appears in the main pane as quickly as the modem and host computer allow. In some older programs, it is necessary to press the reload or refresh button to update to the **latest** message.

Bulletin Boards

Bulletin boards on which learners (both teachers and users) can post messages to be read later by others are built into some CALL environments and more general learning platforms such as **Blackboard Vista**. In such platforms, users can post messages and comment on the original posting and subsequent **comments ad infinitum**

The advantage of bulletin boards over email is that the messages are shared with a **broader community** (a few select people, a class or the whole world) and comment can be more considered as readers have more time. However, in some cases, when two or more people are actively responding to the same posting, communication can resemble a **chat line**.

MOO, MUA, MUD, MUG, MUSH

MOO, MUA, MUD, MUSH and MUG generally refer to the same thing (and are referred to hereafter as **MOO**), an online environment where moveable objects represent things and people. MOO objects may be **photo-realistic 3-D manipulated models or 2-D flat representations**. This environment is a **synchronous** online multi-user space, that is, action takes place in real time among several participants who put their characters into the same scene on a computer screen. These scenes are usually referred to as rooms and can have great visual interest. For example, **'rooms'** can be seaside settings featuring lighthouses and surprisingly active sea life or depictions of castles in the air.

MUDs were originally developed in 1978 by Roy Trubshaw, a student at Essex University, for socializing and playing games online (particularly versions of Dungeons and Dragons). **Richard Bartle** developed the game and helped to promote it internationally. **MOOs** seem a promising direction for further research and materials development although the cognitive overhead of learning how to operate within such an environment may be daunting for some teachers and learners.

World Wide Web Resources [www](#)

Many such websites are driven by commercial concerns. Typically, the learner pays a fee for enrolling and taking online lessons or endures advertising, much of which will be targeted at the learners by, for example, **textbook publishers, language schools and language testing services**. Several publishers also maintain educational websites featuring portions of their work or extra study materials for their work.

Other resources commonly found on the WWW include those created by learners and teachers. These include everything from software to class handouts and presentations in the form of **PowerPoint files**. These resources vary in quality and extent but at least have the virtues of being free and easy to find with a search engine.

Adapting other Materials for CALL

Many other materials on the WWW are easy to adapt for classroom learning and offer a rich source of **authentic** text, images, sound and video. For example, a learner studying **Japanese** can go to countless websites on different topics of interest related to Japan to learn more about culture or information related to Japanese for **Specific** Purposes. Learners can also visit online newspapers for the target language they are studying.

Virtual treasure hunts are activities in which the teacher assigns learners to collect images or explanations on a variety of topics.

Personal Digital Assistants (PDAs) and Mobile Telephones

A **Personal Digital Assistant (PDA)** is a small hand-held computer for downloading and storing information such as documents, databases and calendar entries. It is less powerful than a desktop or laptop computer, but less **expensive** and more portable. **Portability** is achieved by eliminating the keyboard and minimizing the screen size. **PDAs** have largely been replaced in the marketplace with mobile telephones that share many of the same functions and often feature Internet connectivity.

Graffiti Handwriting Software

In **Ray Bradbury's (1958)** collection of **science-fiction short stories**, **The Martian Chronicles**, one tale tells of a man who finds an **alien's deserted** home that miraculously adapts to his needs for water and shelter. In the absence of a **keyboard**, a major challenge to PDAs has been recognition of **handwriting**. Early models struggled to recognize individual handwriting but the creators of the **Graffiti** system realized it would be easier to make users adapt their handwriting to the PDA.

Graffiti used modified and simplified **punctuation, numbers and upper-case letters**. Each character had to be **unconnected**; one could not write on the screen without lifting the stylus between letters. However, each character is a **continuous line**. The system was quickly and easily learned with games that helped develop **fluency**.

CALL and SLA (10. CALL and Second Language Acquisition I. 11. CALL and Second Language Acquisition II. 12. CALL and Second Language Acquisition III)

Second-Language Acquisition (SLA)

Second-Language Acquisition (SLA) refers to the study of the processes through which learners acquire a new language

Negotiation of Meaning

Nunan (1993) offers a concise definition of negotiation of meaning, 'The **interactional** work done by speakers and listeners to ensure they have a common understanding of the ongoing meanings of the discourse' (p. 122). Offering a further explanation of negotiation of meaning, Ellis (1998) provides a brief summary of a theoretical account of how discourse affects language acquisition: 'acquisition is promoted when the input to which learners are exposed is made comprehensible as a result of interactional modifications that arise from a communication breakdown – a process known as **the negotiation** of meaning' (p. 160).

Essentially, if two learners are assigned a so-called **second-language collaborative** task and do not need to communicate, they are often in agreement about how to complete the task and probably do not need to collaborate. That is, the learners are either merely **cooperating** (each completing a part of the task without need for interaction) or the task is so simple that all decisions are obvious.

However, when the task requires that the learners communicate, **grappling with** words, expressions and new ideas that surround the task as well as the technology of the computer's interface and input and display devices, the act of negotiation of meaning may assist in the acquisition of the target language. Of course, negotiation of meaning is not the sole route or complete means for SLA. Pica (1998), for example, notes that learner negotiation of meaning can be quite simple and not cover some important aspects of language learning.

A computer program can provide a high level of **comprehensible input** in various media.

Opportunities for SLA can be offered at the computer when learners are exposed to new language and when learners are prompted to engage in collaboration that promotes negotiation of meaning. Negotiation of meaning and collaboration can occur regardless of the **program; ironically**, some of the worst software programs may present more opportunities for SLA as learners struggle – and discuss their struggles – to make sense of the content and interface.

In terms of CALL, the individualization of instruction makes for even greater opportunities for SLA to be promoted through software designs that assess learners' learning styles and track learners' acquisition through tests which remember and revisit individual items with which each learner has difficulty. However, although this is both possible and desirable, I am not aware of commercial software that does so.

Comprehensible Input and Output

One of the key contentions within SLA is the role of **comprehensible input and comprehensible output**. These terms are explained and discussed in the following section.

Comprehensible input is an idea that originated with **Terrell and Krashen** (Krashen, 1981; Krashen and Terrell, 1983) but several others, including Ellis (1985), have defined it in their own ways.

A.B.M. Tsui on comprehensible output:

Swain (1985) points out that the production of **comprehensible output** is also essential to the acquisition of the target language.

Tsui (1985: 81)

D. Nunan on comprehensible input:

Messages addressed to the learner that, may contain structures and grammar that are beyond the learner's current **competence**, are made understandable by the context in which they are uttered. According to Krashen's **Comprehensible Input Hypothesis**, acquisition occurs when learners' understand messages that are just beyond their current stage of development.

Nunan (1999: 303 – 4)

In terms of CALL, Krashen (cited in Kenning and Kenning, 1990) describes optimal input for acquisition as having four characteristics:

1. It is comprehensible
2. It is interesting and/or relevant to the acquirer
3. It is not grammatically sequenced
4. It is provided in sufficient quantity

CALL can generally deal with all of these criteria, including adding a wide range of **extralinguistic** clues through **sound, images, animation and video**. But a concern with CALL is how the computer ensures that the input for the program is neither too easy nor too difficult for the learner. Such decisions are relatively simple for a well-trained teacher to make, but can be difficult for a computer. **Three** ways in which computer programs deal with this problem are: by offering learner-prompted extra-linguistic elements as clues; by having learners select their own level of comprehension; and/or through

Computer-Adaptive Testing (CAT)

Computer-Adaptive **Testing (CAT)** uses a database of questions to match the difficulty of each test item to the abilities of the learners being tested. Learners take a CAT test at the computer and because the computer can instantly mark each answer, the following question can be tailored or adapted. If a learner answers a question correctly, the computer will ensure that the next question will be more **difficult**. If a learner answers a question incorrectly, the next question will be **easier**.

One of the great advantages of CAT is that **randomization** of test items can ensure that learners of a large class taking a test in the same room may all take slightly different tests as their correct and incorrect answers prompt the computer to take them to different levels. However, it is both **time-consuming and difficult** to set up CAT and learners may not like the fact that they cannot review or change the answers to any questions they have already answered.

P. Lai and J. Biggs on mastery learning:

Mastery learning is based on the assumption that learning is a function of time (Bloom, Hastings and Madaus, 1971; Carroll, 1963). In implementing mastery learning, it is thus necessary to establish a **criterion**, and to provide corrective instruction in the event of failure. Learners failing to reach the objectives initially are given more time in which to pass in **subsequent** attempts. The content to be learned is divided into units, with a **formative** test on each unit.

C.T. Fosnot on mastery learning:

This model assumes that wholes can be broken into parts, that skills can be broken into **subskills**. Learners are diagnosed in terms of deficiencies, called **'needs'**, then taught **until 'mastery'** – defined as behavioral competence – is achieved at each level. Further, it is assumed that if mastery is achieved at each level, then the more general concept, defined by the **accumulation** of the skills, has also been taught. Fosnot (1996: 9)

The challenge of mastery learning in CALL is **the necessity** of providing new material or new approaches when a learner fails to accomplish the initial goals. It is important that, in a restricted time frame, learners do not abrogate the task of thinking and take advantage of a software program's willingness to supply default answers.

Behaviourism takes the view that the learner comes to the learning process with little or no background knowledge. Strict objectives of what is to be learned are broken into instructional steps and rules. Learning activities are sequenced from **simple to complex** with frequent reviews and tests of key points. **Failures** or mistakes lead the learner to repetitions of key parts of the program or **remedial** activities. The control of the sequence or program is usually with the **program**, not the learner.

Constructivism

Constructivism is a humanistic model that differs radically from behaviourism, suggesting that learning is a process by which learners construct new ideas or concepts by making use of their own knowledge and experiences. In a constructivist model, the learner is assumed to come to the class-room with a range **of experiences** and a wealth of (sometimes imperfect) knowledge. It is assumed that knowledge is an **objective interpretation** of ideas and that such interpretations are best developed through the learner discovering and struggling with ideas.

M. Pennington on the components of an ideal teaching system:

- Helps learners develop and elaborate their **increasingly specified cognitive** representation for the second language
- Allows learners to experiment and take risks in a **psychologically** favorable and motivating environment
- Offers input to both **conscious and unconscious** learning processes
- Offers learners opportunities to practice and to receive **feedback on performance**
- Allows learners to learn according to their own **purposes and goals**
- Puts learners in touch with other learners
- Promotes **cultural and social** learning
- Promotes interactivity in learning and communication
- Exposes the learner to appropriate contexts for learning
- Expands the learner's **'zone of proximal development'**

- Builds to learner independence

Schema Theory

Schema theory is important to CALL because it provides an idea of how knowledge is organized. Psychologist F.C. Bartlett (1886 –1969) first proposed the idea of schema theory in 1932. Nunan (1993) defines schema theory as ‘A theory of language processing which suggests that discourse is interpreted with reference to the background knowledge of the reader or listener’ (p. 124). Nunan also notes ‘schema theory suggests that the knowledge we carry around in our heads is organized into interrelated patterns. These are constructed from all our previous experiences and they enable us to make predictions about future experience’ (p. 71).

T.G. Anderson helps to define a schema approach to teaching and learning:

Knowledge is not merely a collection of facts. Although we may be able to memorize isolated facts for a short while . . . meaningful learning demands that we internalize information; we break it down, digest it and locate it in our pre-existing highly complex web of interconnected knowledge and ideas, building fresh links and restructuring old ones.

Anderson (1988) notes that among the differences between experts and novices is the fact that an expert will break down new information into more relevant chunks than a novice will and an expert has more knowledge to which it may be connected. That is, experts will deal more efficiently with new information because their schemata are more developed and, in a sense, more accommodating. Two common ways of creating such schematic maps are word association (e.g. making semantic connections) and pattern noting (examining ideas and the real world and drawing inferences about the relationships).

Example: Mind Map Software

Mind maps are visual representations of schemata. They are usually presented as a series of branching lines radiating from a central circle containing the main idea. The lines terminate in circles of related ideas which may branch off to further ideas. A commercial application that makes creating mind maps easy is Inspiration Software’s Inspiration and a simpler version for young children Kidspiration

Behaviourism

As mentioned earlier in relation to methods and approaches, theories of learning do not appear in a vacuum. They are influenced by earlier practices (e.g. observations of classroom behaviour) and theories from other disciplines (e.g. mathematics, philosophy, psychology). In the case of behaviourism, there is a great deal of earlier practice in teaching methods dating back to the ancient Greeks.

Pavlov’s ideas are now called classical conditioning, as opposed to operant conditioning (explained below). Ideas of stimuli and responses working to modify behaviour were expanded upon and refined by John Broadus Watson (1878 –1958), Edward L. The behaviourist theories of Skinner are based upon the idea that learning is a function of change in overt behaviour. Changes in behaviour are the result of an individual’s response to events (stimuli) that occur in his or her environment. Skinner advanced previous behaviourist theories by showing how a response produces a consequence such as defining a word, hitting a ball or solving a Maths problem. When a particular stimulus and response pattern is reinforced through

rewards, the individual has been conditioned to respond. Skinner called this approach operant conditioning and it differs from earlier forms of behaviourism (now called classical conditioning) in that it recognizes that a person (or animal) can emit responses and not only elicit responses in reaction to a stimulus.

C.T. Fosnot on ways in which educators incorporate behaviourism in the classroom:

In terms of a practical application to classroom teaching and learning, as well as CALL, behaviourism was developed into two methods: programmed instruction and mastery learning. These are discussed below before moving on to a discussion of constructivism and a comparison of behaviourism and constructivism.

Programmed Instruction

One of the practical applications of the behaviourist approach is the design of programmed instruction or programmed learning on which Skinner's behaviourist contributions to CALL centre.

The Role of Collaboration and Negotiation of Meaning in the Two Models

Collaboration and negotiation of meaning are dealt with in greater detail in this lesson but it would seem, on surface examination, that the constructs of collaboration and negotiation of meaning fall solely within the constructivist model. The implication is that cooperation/collaboration and negotiation of meaning may exist as features of both behaviourist and constructivist models of instruction. However, the degree to which they are effective within each model needs to be examined.

CALL Evaluation (13. Principles for CALL Evaluation. 14. Judgmental Evaluation of CALL. 15. Empirical Evaluation of CALL 16. Research Methods in CALL)

Summary of principles for evaluating CALL

Principle Implication Evaluation of CALL is a situation- Specific argument. CALL developers need to be familiar with criteria for evaluation which should be applied relative to a Particular context.

CALL should be evaluated through two perspectives: judgmental analysis of software and planned tasks, and empirical analysis of learners' performance.

Methodologies for both types of analyses are needed

Criteria for CALL task quality should come from theory and research on instructed SLA.

CALL evaluators need to keep up with and make links to research on instructed SLA.

Criteria should be applied in view of the purpose of the task.

CALL tasks should have a clearly articulated purpose.

Language learning potential should be the central criterion in evaluation of CALL.

Language learning should be one aspect of the purpose of CALL tasks.

Judgmental and Empirical Analyses

The first level of **analysis** refers to the software that is used for a CALL activity. The second level of analysis is directed toward the **teacher's planned activity**. The third level of evaluation focuses on **learners' performance**, and is therefore conducted through examination of empirical data reflecting learners' use of CALL and learning outcomes.

An evaluation argument should be constructed on the basis of both the **judgemental and empirical** analyses. These two methods provide different and complementary information both of which are relevant to CALL task evaluation (Chapelle, 1999a). The **judgemental analysis** should examine characteristics of the software and task in terms of criteria drawn from research on SLA. The **empirical analyses** address the same criteria but through data gathered to reveal the details of CALL use and learning outcomes.

Criteria from Theory and Research on SLA

Drawing from the theory and research on conditions for instructed SLA in addition to the other considerations mentioned above (e.g., individual differences), it is possible to chart some criteria for evaluating CALL.

Language learning

The degree of opportunity present for beneficial focus on potential form.

Learner fit

The amount of opportunity for engagement with language under appropriate conditions given learner characteristics.

Meaning focus

The extent to which learners' attention is directed toward the meaning of the language.

Authenticity

The degree of correspondence between the CALL activity and target language activities of interest to learners out of the classroom.

Positive impact

The positive effects of the CALL activity on those who participate in it.

Practicality

The adequacy of resources to support the use of the CALL activity.

Learner Fit

Whereas language learning potential captures the findings concerning general processes, **learner fit** takes into account the individual differences in linguistic ability level and non-linguistic characteristics.

Learner characteristics such as willingness to communicate, age, and learning style also come into play in task choice.

Meaning Focus

Meaning focus denotes that the learner's primary attention is directed toward the meaning of the language that is required to accomplish the task, the clearest example being communication tasks as defined by Pica, Kanagy, and Falodun (1993). Their primary defining feature is that they require learners to use the target language to accomplish something such as making a decision on an issue, or exchanging information to accomplish a **goal**. Such tasks differ from **form-based** tasks which might have learners filling in correct verb tenses in a written list of sentences, or changing **declarative** statements to yes/no questions in an oral drill. Meaning focus is not limited to **oral communication tasks**, but also can occur during tasks involving reading and writing when learners use the written language purposefully for constructing and **interpreting** meaning.

Authenticity

The criterion of **authenticity** indicates the need to develop learners' willingness to communicate but it also extends beyond the conditions believed important for acquisition. Authenticity refers to the degree of correspondence between an L2 learning task and tasks that the learner is likely to encounter **outside** the classroom. The choice of **pedagogical tasks** that learners see as relevant to their language use beyond the classroom should help to engage learners' interest and therefore their willingness to participate. Moreover, current theory of communicative language ability (Bachman, 1990; Bachman & Palmer, 1996) defines it as **situation specific**, implying that development of ability in language for particular purposes requires practice in using language for those purposes.

Positive Impact

The positive impact of a CALL task refers to its effects beyond its **language learning** potential. The significance of this quality has been pointed out for **assessment tasks** (e.g., Bachman & Palmer, 1996), but it is equally important for learning tasks. **Ideally** classroom language learning tasks teach more than language;

Practicality

Practicality refers to how easy it is for the learners and teachers to implement a CALL task within the particular constraints of a class or language program. **Relevant constraints** include the availability of hardware and software that are adequate for the planned activities. In addition, **knowledgeable personnel** need to be on hand to assist with **unforeseen** problems. Early experience with CALL showed that learners had to have **adequate** access to well-maintained software and hardware for CALL to be successful (Marty, 1981). This observation is equally valid today because even though learners use computers regardless of **infrastructure** provided by language pro-grams, they cannot be expected to use computers for language learning without guidance, and guidance requires resources. Issues of practicality are closely tied to characteristics of **institutional, social, and cultural practices** in which some members have the power to make decisions about the amount and type of resources to be made available for CALL.

Criteria Applied Based on Task Purpose

These criteria for CALL **appropriateness** need to be applied in view of the purpose of a CALL task. Skehan's discussion of performance goals for tasks, including **learners' fluency, accuracy, and complexity**, might be augmented by comprehension goals and goals concerning pragmatic competence (Kasper, 1997). Moreover, tasks may have different purposes at various stages of instruction (Doughty & Williams, 1998). Whatever the goal of the CALL task, however, evaluation of the **task** requires that it have a stated purpose.

Language learning

Do task conditions present sufficient opportunity for beneficial focus on form?

potential

Learner fit

Is the difficulty level of the targeted linguistic forms appropriate for the learners to increase their language ability?

Is the task appropriate for learners with the characteristics of the intended learners?

Is learners' attention directed primarily toward the meaning of the language?

Meaning focus

Authenticity

Is there a strong correspondence between the CALL task and second language tasks of interest to learners outside the classroom?

Will learners be able to see the connection between the CALL task and tasks outside the classroom?

Impact

Will learners learn more about the target language and about strategies for language learning through the use of the task?

Will instructors observe sound second language pedagogical practices by using the task?

Will both learners and teachers have a positive learning experience with technology through the use of the task?

Practicality

Are hardware, software, and personnel resources sufficient to allow the CALL task to succeed?

Computer-Assisted Classroom Discussion

Kelm (1992) provides an example of a **LAN-based computer-mediated communication (CMC)** activity in a university-level, **fourth-semester** Brazilian Portuguese class held in the US. Students attended class three hours a week, and for one of those hours (**each Friday**), class was held in the microcomputer center

of the university library. Before coming to class each Friday, they were assigned a particular Brazilian short story which was to serve as the topic for the **computerassisted classroom discussion (CACD)**.

Micro-world

Chun and Brandl (1992) described a **micro-world** activity designed for beginning learners of **German**: 'The **functional** goal of this situation is [for the learner] to locate objects in a room and to differentiate between **stationary physical location vs. the action or motion involved in placing an object somewhere**' (Chun & Brandl, 1992: 260).

Text Analysis

Liou (1993) described the use of a grammar checking program in a **first-year writing and grammar course for EFL** majors at a university in Taiwan. The writing class took a processoriented approach in which learners were required to write, participate in **peer editing, receive comments from the instructor, and revise**.

Storyboard

The **storyboard** activity **Jones and Fortescue (1987)** described was used by a group of students who sat together in front of a computer screen in a computer lab. 'The text is entirely obliterated, and the learners can see only the title (**Superstition**), a mass of blobs, a reference to various help features and an invitation to guess a word' (1987: 37). The activity is therefore a **guessing game**, which is set up as a storyboard containing a text on superstition. The learners work **collaboratively** through oral conversation to determine what, when, and how they will input words into the game. The conversation among the learners **may or may not be** conducted in the target language, and the learners are free to take their time producing the language used in the reconstruction.

Empirical Evaluation of CALL

Empirical evaluation offers a methodology for making systematic hypotheses about the benefits to be attained through CALL tasks.

Evidence for CALL Qualities

Empirical research methods for **evaluating** L2 classroom tasks have to a large extent given up on evaluating language instruction solely through measurement of learning outcomes in favor of **investigating** classroom processes.

Research Methods

The most convincing way to demonstrate the language learning potential of a CALL activity is through the study of **learning out-comes**.

Comparing CALL with non-CALL Activities

CALL research attempting such discourse analysis has examined learners' **oral language** as they worked on CALL programs (e.g., Abraham & Liou, 1991), and findings have been mixed.

Positive Impact

Descriptions of CALL throughout the past 30 years abound with statements about the positive influence of CALL activities on language classrooms. One can pick up virtually any issue of any CALL journal from the past 15 years to find examples of enthusiastic CALL users' estimations of the experience of CALL as positive for their students. The current generation of CMC enthusiasts is no less euphoric about the impact of their CALL tasks, arguing that such activities offer the positive impact of changing classroom dynamics (Beauvois, 1992; Collombet-Sankey, 1997; Swaffar et al., 1998).

Practicality

Questions about practicality of CALL rely on evidence suggesting that hardware, software, and personnel resources prove to be sufficient to allow CALL to succeed.

Theoretical and Pedagogical Concerns (17. Theoretical and Pedagogical Concerns in CALL- I. 18. Theoretical and Pedagogical Concerns in CALL- II. 19. Theoretical and Pedagogical Concerns in CALL- III)

Computers, software and WWW-based services are not seen solely as positive agents of change in the classroom; they also face criticism. In the already mentioned example of young and second-language learners lacking dictionary and library search skills, a multimedia resource may prove to be too seductive an information source. The result is a delay or even complete omission of their learning of traditional and useful information search skills, just as a reliance on electronic calculators has discouraged some learners' basic numeracy skills.

Concerns for software development

As already mentioned, traditional materials tend to follow a set scope and sequence that lay down the paths and principles of learning and this is the general route followed by most computer-based learning materials that are, in some cases, adaptations of existing textbooks. However, a strength – and a weakness – of some computer-based materials is their lack of a clearly determined scope and sequence. Instead, they allow individual learners to pursue links which they perceive as being both useful and interesting.

In collaborative situations at the computer, a good interface would be one that prompted learners to share control. This is often done in game software and some web applications that require two or more users to register their names. These names can be used as prompts for computer-directed actions and turn taking.

Software Objectives

Language teachers generally aim to increase learner-Centred ness, or locus of control towards the user, in activities in which, as Nunan (1984) suggests, learners can be more closely involved in the decision-making process of what is taught and how it is taught. Creating a schema, or mental map, of learners' needs along with notes on the software packages which match each need is a good way to allow learners to organize their own learning. Such a chart could also include an overview of the curricular objectives to indicate how learners should progress through levels of language learning. This can involve learners in the process and make them much more responsible for their learning and, in doing so, also increase intrinsic motivation.

Lack of Funds

The previously mentioned **Athena Language-Learning Project (ALLP)** project was singular in attracting funding in excess of **US\$70 million**. Few commercial organizations spend even a fraction of this amount on software development and non-commercial programs are written on far smaller budgets or, more commonly, on no budget at all. Instead, many teachers create software with their own time and resources. In such cases, authors may act conservatively as they are wary of not achieving a return on their investment of time and money. However, since the ALLP was set up many of the costs of producing quality multimedia have been dramatically reduced. For example, computers have become exponentially more powerful while costs have continued to drop. Now authoring tools and related programs for the creation of graphics, animation and video, which were previously developed from scratch, are commercially available at reasonable prices. However, in an endless cycle, a parallel consumer (e.g. learner) demand for greater sophistication in learning materials usually follows each new capability of these authoring programs so underfunded developers may find it difficult to interest learners.

Lack of Expertise

Those creating CALL software programs are often experts in computer programming, design or pedagogy, but are seldom experts in all three fields; while one aspect in a finished program may shine, others may be problematical. Authoring professional software is a labor-intensive enterprise employing the skills of many different people including materials writers, content editors, graphic designers, sound designers, musicians, voice actors, marketers, animators, videographers and so on. A single teacher or group of learners may be disappointed with their efforts when compared with professional products. One solution in a school context is to assemble a team of experts (or at least people skilled in one or more areas), but even within academic institutions this is not always practical and requires both resolution and organizational skills. This problem is particularly severe in the case of individuals trying to create a small program to solve a local task. A second solution is liaison with commercial publishers, who might help edit and develop locally relevant software packages – as long as doing so does not shift the focus and defeat the purpose of developing learning materials for the local market. Example 8.2 Online collaborative projects Creating ‘Call for Participation’ projects Unless you plan to do the project on your own, you’ll need some partners. Before you ask people to join your project, you need to do some planning. First, create a ‘Call for Participation’ that provides potential participants with an overview of your project. Second, develop the materials that you and your participants will need for the project such as lesson plans and project guidelines.

Lack of suitable authoring programs

Many authoring programs are limited in the **question types and information-collection systems** that they offer. **Authoring** programs for CALL and other subjects most commonly include **low-level** question types such as true/false and multiple choice questions as opposed to higher-level synthetic, analytical and evaluative tasks. **Developers** tend to use what is available and lack either the skills or the inclination to develop new authoring tools. Conversely, new software features are often included regardless of whether or not they are **pedagogically** appropriate. This latter problem is less easy to address, but new software is slowly being developed that is based on competition and consumer demand for improved **interactivity, intuitiveness and other features**.

Commercial Software

Comprehensive computer-based learning materials have in some cases been taken out of the hands of educators. Instead, the **cost involved** in the creation of interactive educational software means that it is often designed on the basis of what engineers decide is possible and what marketing executives decide can be sold.

Making better use of existing materials

Ravitch (1987) reviews the history of technology in the school, including film, radio and television, with some pessimism, suggesting that factors peripheral to the technology itself (organization, cost, schedules) have resulted in repeated failures. She also wonders whether new learning materials will devalue **printed materials** and wonders whether technology's instant gratification will dampen learners' abilities to analyze materials' arguing that in 'literature, for example, certain **genres** were meant to be read, not performed. Novels, short stories, essays and poems were meant to be read (p. 32). However, the **idea that literary forms should** only be appreciated in their original medium is **not generally shared**. For example, few have heard Homer's Iliad sung aloud as the author 'intended' and the plays **of Shakespeare** have been not only staged, but also widely read and reinterpreted in art, opera and film. **Multimedia** can be used simply as a new tool for examining old forms as well as for creating new ones, such as **hypertext** novels. It is important for learners to be involved in using **computers** as tools to activate their knowledge by **transforming** their learning into new media.

Writing the hypertext novel

The **non-linear hypertext novel in which the story changes depending on which links one chooses to follow** seemed like an attractive alternative to printed literature, but the form has failed, so far, to attract much interest beyond the art world. A **pioneer** in this field is Michael Joyce who, in 1987, developed the first example, publishing it in 1991.

Copyright and plagiarism

All computer-based information, whether text, graphics, music or software programs, is essentially **digital** in nature. **Digital** means that the core data can be broken down into strings of binary numbers: **ones and zeros**. The **practical side** of this is that much of what we see on the computer can be **copied and manipulated**. This is especially true of text and images found on the **WWW**. Student projects often **feature text and images borrowed** from existing websites. In many countries, a fair use provision within **copyright** law allows for learners to use some materials for **in-class projects**. However, it does not give learners the right to repost images and text onto the **WWW**. **Plagiarism** using materials from the WWW is also common but tools such as those found at www.plagiarism.org can be of some help to teachers who suspect their students of failing to acknowledge what they have borrowed.

The WWW presents **unparalleled** opportunities for plagiarizing materials but it also presents some defence. The simplest tool a teacher has to check for plagiarism is to type a **string of suspect** text into a search engine and see whether it leads to one or more websites from which the suspect text may have been taken.

Turnitin <http://writecheck.turnitin.com> – This website essentially automates the above process but additionally, each document that is submitted for plagiarism checking is added to the database.

Viruses

Viruses are distributed through the WWW and email and can destroy files. There are **countless** viruses in **circulation** with new ones being added each day. Some viruses end up costing computer users a fortune simply by **slowing performance** of computers. Others destroy data. It is difficult and sometimes expensive to maintain **anti-virus** software to handle the latest versions of viruses which, almost by definition, are designed to thwart existing anti-virus programs.

A virus is a **program** and, in most cases, must be activated by the user to make it work. **Traditionally** viruses carried an **.exe file** name, that is, executable files, but they now have various other suffixes or are contained in files with innocuous .doc or .jpg file formats. A common way to receive such files is through **unsolicited** email or through the purchase of illegal software. Once a virus enters your computer, it will often spread itself by taking advantage of the address list in an email program, sending a copy of itself with an email to each name on a user's **address list**. A certain amount of what is called '**social engineering**' is used to make such viruses successful.

Thought viruses

Misinformation is one of the greatest challenges on the WWW and in many ways constitutes a form of thought virus in which erroneous information is passed on from one host reader to another, often through redistribution of email messages.

Safety online

A popular New Yorker cartoon by **Peter Steiner** shows a dog sitting at a keyboard with the caption 'On the internet, no one knows you're a dog' (http://en.wikipedia.org/wiki/File:Internet_dog.jpg). The cartoon reflects the fact that some people may assume new **personae, age and even a different gender** online. A sad consequence of this is that, on unmoderated chat lines, learners may encounter **pedophiles** posing as children or even older learners asking inappropriate questions and using **inappropriate** language.

Censorship

Governments have used censorship to control some of the negative aspects of the WWW, for example by shutting down **Internet Service Providers (ISP)** who carry negative content or content they do not like. Products such as **CYBERSitter** are aimed at children while other products such as **Hate Filter** are aimed at helping adult audiences. However, such programs also have their **drawbacks**.

Cyberbullying

Despite frequent mention in the press, often in connection with the suicide of a bullying victim, little attention is paid in the literature to the prevalence and impact of **cyberbullying**, a term used to encompass any kind of online harassment, including threatening emails and postings to social networking pages like **MySpace**.

Technological have-nots

S. Berger on the digital divide between developed and developing countries:

It is this disparity between the 'technology-rich' and 'technology-poor', or 'have-nots', that is commonly referred to as the international digital divide. Industrialised countries are home to 88% of all internet users, yet make up only 15% of the world's population. Unsurprisingly, PC ownership levels differ dramatically between developed and developing nations. For example, in South Asia only 4 persons per 1,000 own a PC compared to the 585 per 1,000 in the US.